HelioVolt System Architecture Overview

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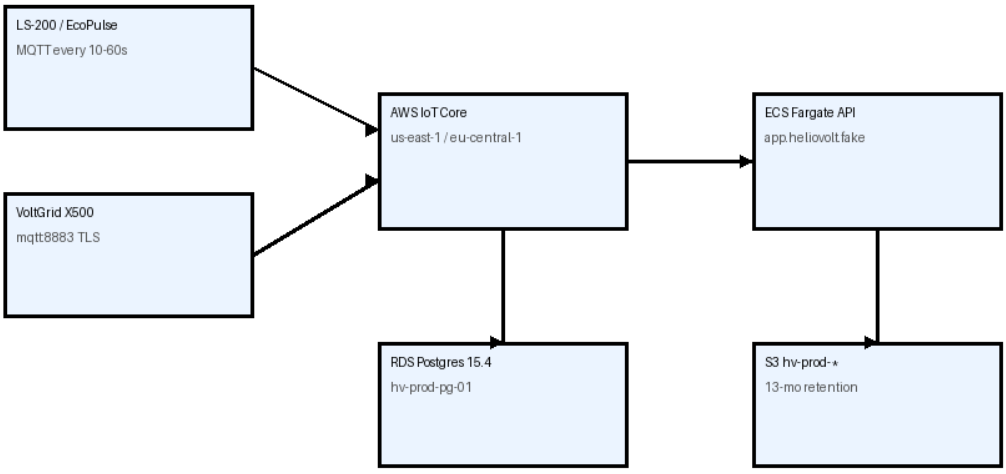

Fig 1: Devices -> IoT Core -> Fargate -> RDS/S3. Secrets in hv-prod/*, Okta SSO, Datadog + Pagers.

Figure 1: End-to-end flow. Devices (EcoPulse, VoltGrid X500, LS-200) publish to MQTT broker mqtt.heliovolt.fake:8883 via TLS. AWS IoT Core routes to ECS Fargate API (app.heliovolt.fake US, app-eu.heliovolt.fake EU in eu-central-1, VPC vpc-eu-042). Hot data in RDS Postgres 15.4 instance hv-prod-pg-01, cold telemetry in S3 hv-prod-telemetry (13-month rolling, then aggregated). Secrets in AWS Secrets Manager hv-prod/*. Auth via Okta SSO (Okta Verify, no SMS). Observability: Datadog + PagerDuty + Slack #eng-alerts. SEV1 if down >5 min or error >5%.

Key Decisions: EU data never leaves eu-central-1 (GDPR). Rate limits 600 req/min Professional, 2000 Enterprise. retains CloudWatch 90 days. Deploy freeze Fri 3pm-Mon 9am CT; rollback SLA 15 min via rollback.yml to prior version (e.g., v4.1.7). Status at status.heliovolt.fake.